



PROFESSIONAL CERTIFICATION

AutoCAD Civil 3D Course Training

Comprehensive AutoCAD Civil 3D course focused on infrastructure design as per UAE Dubai Municipality and Abu Dhabi standards. Learn surface creation alignments profiles assemblies subassemblies road corridors section views gravity and pressure networks bridge modeling BIM parameters for roads utilities bridges and rail projects.



35-50

Hours



Certified

Trainers



Industry

Expert



Professional Training | Industry-Recognized Certificate

Dubai, UAE

About Orbit Training Center

Your Gateway to Professional Excellence in Dubai



Who We Are

Established in 2023, Orbit Training Center is a leading professional training provider in Dubai, offering 60+ professional certification courses across diverse fields, including engineering, IT, design, and languages.



Our Benefits

-  Industry recognized certifications
-  Convenient location in the heart of Dubai
-  Affordable fees with multiple payment plans
-  Lifetime post-training support

★Our Features

-  Hands-on learning with real-world projects
-  Expert trainers with industry experience
-  Flexible learning options to fit your schedule
-  State-of-the-art infrastructure & facilities



Our Mission

We deliver outstanding courses, expert teaching, strong career guidance, and placement support to help every student achieve their best results. We stay connected with alumni through networking events, lifelong learning, and ongoing mentorship. We run our operations with smart systems and data-driven decisions to create a smooth, exceptional experience for all.

AutoCAD Civil 3D Course

Practical AutoCAD Civil 3D training for road bridge utility design surfaces corridors networks and UAE municipal BIM standards.

Course Description

This course delivers hands-on AutoCAD Civil 3D skills tailored for UAE infrastructure projects including Dubai and Abu Dhabi requirements. Cover surface modeling alignments profiles assemblies corridor design sections gravity pressure networks bridge elements BIM parameters earthwork calculations and compliant drawings for roads utilities bridges rail.

Duration

35-50

Hours of Training

Format

Classroom / Online

Flexible Option

Key Outcomes

- ✓ Create intelligent surfaces from points data
- ✓ Design alignments profiles cross sections
- ✓ Build road corridors assemblies targets
- ✓ Model gravity pressure pipe networks
- ✓ Develop bridge components alignments
- ✓ Generate section views quantity takeoffs
- ✓ Produce UAE-compliant BIM drawings reports

Target Audience

- Infrastructure designers draftsmen UAE projects
- Road bridge utility engineers Dubai Abu Dhabi
- Students mastering surfaces alignments corridors

Certification

- Certificate of Course Completion
- Industry Recognized Certification
- Trained by Certified Professionals

Prerequisites

- Basics AutoCAD commands drafting required
- Road geometry civil engineering knowledge essential
- Coordinate systems units familiarity needed

AutoCAD Civil 3D Course

Start AutoCAD Civil 3D with surfaces alignments corridors networks bridge design BIM UAE standards.

Intermediate

- Covers all Intermediate Levels
- Precise Geometry Creation
- Basic Parameters & Standards
- Understanding of Tools & Uses
- Intermediate Project Standard
- 1 Project at Completion

Duration
35 Hours

Professional

- Covers all Professional Modules
- 16 Standard Professional Modules
- Learn with Industry Standard
- Professional Tools and Uses
- Multiple Professional Projects
- 3 Projects at Completion

Duration
43 Hours

Advanced (Customized)

- Covers All Professional Modules
- 16+ Advance Professional Module
- Co-Ordination and Adv. Parameters
- Shop Drawings & As Built Drawings
- Govt. Standard and Submission
- 4 Projects in Different Categories

Duration
50 Hours

AutoCAD Civil 3D Course

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01 Interface Navigation and Project Setup Fundamentals

- › Customize ribbon tabs panels and tool palettes for productivity
- › Configure units scales coordinate systems and drawing templates
- › Set up annotation styles layer standards and CAD checking protocols
- › Optimize command line dynamic input and viewport navigation controls
- › Implement backup recovery and file management protocols

02 Coordinate Systems and Geospatial Data Management

- › Assign accurate coordinate reference systems for infrastructure projects
- › Georeference drawings using control points survey metadata and basemaps
- › Perform coordinate transformations datum adjustments and validation
- › Integrate aerial imagery topographic maps and contextual data layers
- › Validate geolocation accuracy and maintain traceability for compliance

03 Survey Points Import and COGO Point Management

- › Import points from various survey data formats and field equipment
- › Create and edit dynamic COGO point groups with description keys
- › Apply point styles labels and automate point classification
- › Perform data reduction outlier filtering and quality checks
- › Export points for construction staking and field verification workflows

04 Terrain Surface Creation and Editing Techniques

- › Build TIN surfaces from survey points breaklines and boundaries
- › Define surface boundaries edit definition order and manage data
- › Apply smoothing simplification and contour generation techniques
- › Generate contour labels elevation banding and surface analysis displays
- › Calculate accurate cut-fill volumes between existing and design surfaces

05 Horizontal Alignment Design and Geometry Control

- › Create alignments using PI tangent curve and layout methods
- › Edit geometry with grips parameter controls and design criteria
- › Apply design criteria files and check AASHTO or local standards
- › Generate station equations geometry tables and alignment reports
- › Produce offset alignments transition widenings and intersection geometry

06 Vertical Profile Design and Profile View Configuration

- › Sample existing surfaces to create ground line profile geometry
- › Design vertical curves with parabolic circular and asymmetric types
- › Edit profile entities using precise parameter and grip controls
- › Customize profile view grids band sets and data annotations
- › Perform sight distance K-value calculations and design validation

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07 Parametric Assembly and Subassembly Development

- > Assemble conditional subassemblies for flexible corridor designs
- > Configure targets for width slope elevation and surface overrides
- > Create custom subassemblies using Subassembly Composer tool
- > Manage version-controlled assembly libraries for project standards
- > Visualize subassembly behavior and validate in section editor

08 Road Corridor Creation and Advanced Editing

- > Build corridors with multiple baselines regions and frequency settings
- > Apply targets frequency and surface overrides for complex designs
- > Generate corridor surfaces feature lines and boundary extraction
- > Edit corridors interactively in 3D views and cross sections
- > Audit optimize and troubleshoot corridor model performance issues

09 Advanced Grading Techniques and Site Design

- > Create feature lines with surface elevation and grading assignments
- > Design grading objects using slope projection and distance rules
- > Manage grading groups infill surfaces and volume calculations
- > Produce volume surface comparisons and earthwork reports
- > Integrate grading with corridor daylight transitions and transitions

10 Parcel Layout and Subdivision Design Management

- > Generate parcels from alignments boundary lines and slide lines
- > Apply parcel sizing swingline editing and area optimization tools
- > Configure parcel styles property data tables and area labels
- > Produce area reports ownership summaries and legal descriptions
- > Perform right-of-way easement and setback calculations accurately

11 Gravity and Pressure Pipe Network Design

- > Layout gravity sanitary storm and pressure water pipe networks
- > Configure design rules for cover slope and hydraulic requirements
- > Edit structures fittings and customize parts catalog items
- > Generate pipe profiles and perform interference clash detection
- > Produce annotated pipe network plan and profile drawings

12 Railway Alignment and Corridor Modeling

- > Create rail alignments with cant transition and spiral geometry
- > Apply superelevation rail design criteria and speed parameters
- > Build rail assemblies including ballast sub-ballast and track layers
- > Generate rail corridor surfaces and clearance envelope analysis
- > Produce cross sections cant diagrams and rail-specific reports

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13 Bridge Approach and Structural Component Modeling

- › Develop bridge approach alignments profiles and transition geometry
- › Assemble parametric deck girder and abutment subassemblies
- › Model transitions using conditional target logic and variations
- › Generate bridge corridor surfaces and structural quantity takeoffs
- › Create annotated bridge plan elevation and section sheet drawings

14 Professional Annotation Labeling and Table Creation

- › Apply dynamic labels to alignments profiles and Civil 3D objects
- › Customize label styles for plan profile and section drawings
- › Create label sets for consistent annotation across project sheets
- › Control label masking precision drag state and visibility settings
- › Produce geometry tables volume summaries and quantity reports

15 Survey Database Management and Field Integration

- › Import and process large survey field book databases efficiently
- › Automate figure creation with linework code sets and standards
- › Edit survey points observations and network adjustments accurately
- › Link survey data dynamically to design surfaces and corridors
- › Validate survey integrity for construction handover and as-builts

16 Data Shortcuts Team Collaboration and Project Management

- › Establish production data shortcut project workflows and structures
- › Synchronize and promote referenced Civil 3D objects across drawings
- › Resolve reference conflicts broken paths and synchronization issues
- › Configure secure folders for multi-user team environments
- › Implement best practices for large team projects and coordination

17 Quantity Takeoff Mass Haul and Cost Analysis

- › Define material criteria and volume calculation lists accurately
- › Compute pavement subgrade and earthwork quantities precisely
- › Generate mass haul diagrams balance lines and haul reports
- › Compare surface scenarios for optimization and value engineering
- › Export quantity data to external estimation and bidding software

18 Construction Documentation and UAE BIM Compliance

- › Create annotated plan-profile and section sheet sets professionally
- › Configure sheet set manager publishing and plotting workflows
- › Export models to IFC LandXML and COBie formats for BIM delivery
- › Align deliverables with Dubai Municipality and Abu Dhabi DMT standards
- › Ensure compliance with UAE permit sustainability and Vision 2030 rules

Student Google Reviews

Hear What Our Graduates Say About Their Experience



Muppuli Raja



One of great institute for REVIT BIM course. I completed my training with Mr Ahmed. He is one great trainer in this institute. He has very good knowledge for BIM Training. I completed my REVIT BIM training with orbit and it was very nice learning journey for me. And one of best thing is training center is very flexible for classes.



Arun



Best SolidWorks training in Dubai! The instructor made 3D modeling and assembly design easy to understand. The course covered all essential tools I need for product design and engineering work. Best training center for solidworks training. Best Training Center for Best Skills



Harshu CH



My experience at Orbit training center has been the best In every step I learn from there very clearly almost complete my course it's very interesting the trainer is the best professional and friendly ,and the rest of the team keep it up, thanks for supporting me Orbit and all staff...



Ranjan Axalan



I just finished my course in 3ds max. I learned a lot on how to create 3d model. Thank you for the experience and for the teaching method it help me to understand easily and grow my knowledge. I will use my learnings to my future company. Highly Recomanded!



Numan



I had good support from trainer Mr Ahmed and the other staff. They would schedule my BIM classes as per my convenience and would retake classes if I required. They always answer my calls when I need assistance. Not only that the trainer would go a mile a ahead by helping me with my work related issues like a true mentor.



Lelanie Fernandez



Overall, I highly recommend the Fusion 360 training at Orbit Center, especially with Ahmad as the instructor. This course has exceeded my expectations, and I am learning a lot. I would rate it 5 out of 5 stars!

Student Google Reviews

Hear What Our Graduates Say About Their Experience



Aman Bakhtiyar



I had a great learning experience at Orbit Training Center. The trainers are knowledgeable, supportive, and always ready to help. The clear and practical teaching methods really helped improve my skills and confidence. I highly recommend Orbit Training Center to anyone looking to learn AutoCAD, Revit, and other professional skills.



Sana Zahid -



I recently completed the online Digital Marketing course with Orbit Training Center, and it was a great experience! The lessons were clear, practical, and easy to follow. The trainers were knowledgeable and supportive throughout the course. I've learned a lot that I can apply to my work here in Belgium. Highly recommend Orbit Training Center to anyone looking to improve their digital marketing skills!



Cleavon Henderson



Orbit Training Center offers outstanding CAD classes that are easy to understand, even as a first-year mechanical engineering student. The instructors simplify complex concepts and use practical examples, making the learning process engaging and effective. Whether you're new to CAD or improving your skills, Orbit Training Center provides a supportive and enriching environment. Highly recommended!



Danish Khan



Just finished Primavera P6 training at Orbit Training Center and it was fantastic! The trainer made complex scheduling concepts easy to understand with real-world examples. Got hands-on experience with project planning, resource allocation, and progress tracking. The course was well-structured and I feel confident using Primavera P6 for my projects now. Great value and would definitely recommend to other project managers!



500+ Students Trained

Join our growing community of professionals

95%

Satisfaction Rate

60+

Courses Offered

4.6

Average Rating



GET STARTED TODAY

Get in Touch with Us

At Orbit Training Center, we offer a diverse range of professional courses designed to equip you with the latest industry skills and knowledge. Our comprehensive programs cover in-demand fields such as Interior Design, Computer Science, Graphic Design, Digital Marketing, Solar Energy, and specialized software training including STAAD Pro, Revit Architecture, Tekla Structures, and more. Each course is crafted by industry experts to provide hands-on learning, real-world applications, and certification preparation.

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Free

Demo Session



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